

Equilibrium-Driven Vertical Federated Learning with Selective Privacy Protection

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Abstract

Vertical Federated Learning (VFL) enables multiple clients with feature-partitioned data to collaboratively train models while preserving privacy by transmitting embeddings instead of raw data. However, such embeddings can still expose sensitive attributes (e.g., gender or race) unrelated to the target task, making them vulnerable to attribute inference attacks. Most existing privacy strategies may provide extra protection, but at the cost of reduced accuracy and excessive privacy budget. In this paper, we propose a novel equilibrium-driven VFL framework with selective privacy protection for sensitive attributes that are difficult to isolate from embeddings, thereby enhancing local privacy with minor accuracy compromise. We introduce two key innovations: (1) a *NashCoder*, which incorporates a surrogate head to jointly optimize accuracy and privacy; (2) an adaptive decomposition strategy based on Shapley values, which dynamically decomposes the global objective for distributed optimization from an equilibrium perspective. We theoretically analyze our framework and empirically evaluate it on three public datasets against five baselines, demonstrating significant improvements in the accuracy-privacy trade-off under various privacy settings. Extensive experimental results support our theoretical analysis.

1 Introduction

Federated Learning (FL) is a distributed learning paradigm that enables multiple clients to collaboratively train models while preserving data privacy. FL can be categorized into two main types: Horizontal Federated Learning (HFL) and Vertical Federated Learning (VFL). In HFL, data is horizontally partitioned across different clients, with each client holding a different sample space but sharing the same feature space. In contrast, VFL involves vertical partitioning across different clients, with each client holding a different feature space but sharing the same sample space. Specifically, clients with distinct but complementary feature spaces collaboratively train models for mutual benefit. For instance, in healthcare, different hospitals hold various medical features for the same patients and collaborate on diagnostics. In multimodal scenarios, different modalities for the same samples are held by distinct clients (e.g., cloud service providers) due to privacy concerns (Peng et al. 2024), while they collaborate on a global target task, as shown in Fig. 1 (a). VFL ini-

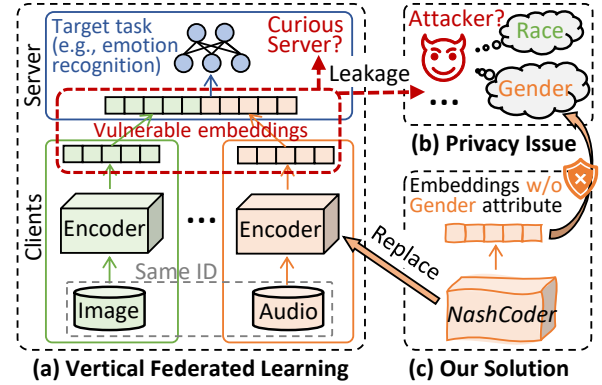


Figure 1: We aim to (1) design *NashCoder* to obfuscate sensitive attributes, and (2) balance the trade-off between global accuracy and local privacy from an equilibrium perspective.

tially protects privacy by (1) splitting the feature space so that each client holds only a partial dataset, and (2) training local models that map raw data into meaningful representations (embeddings) for transmission.

However, an honest-but-curious server or a potential attacker may gain access to these embeddings and maliciously infer sensitive attributes (e.g., gender or race) that are unrelated to the target task by feeding the embeddings into an adversarial classifier. This type of threat is known as an attribute inference attack (Liu et al. 2022a) and is the focus of this work. Notably, the sensitive attributes are commonly predefined and privately known to each client when aiming to enhance privacy. However, they are not in the form of features that can be easily isolated or removed, and are inevitably reflected in the embeddings to some extent for two reasons. First, low-level features are entangled, e.g., the target task (emotion) prediction can be informally expressed as $\hat{y} = f(x)$, while the sensitive attribute (gender) prediction is $\hat{z} = g(x)$, both based on the input x . Thus, sensitive attributes cannot be easily removed by dropping a few selected dimensions from x , as one might do by deleting specific private columns from a CSV file. Second, $g(\cdot)$ is unknown in practice, making it nearly impossible to isolate the sensitive attribute. Thus, embeddings inherently carry privacy leakage risks and are vulnerable to attribute inference attacks.

Most existing privacy-enhancing methods in VFL, such as Differential Privacy (DP) (Tran et al. 2023), Homomorphic Encryption (HE) (Gong et al. 2023), and Secure Multi-client Computation (MPC) (Huang et al. 2023; Li, Yao, and Liu 2023), provide additional protection, but often at the cost of degraded target task accuracy and excessive privacy overhead, as they fail to identify which attributes are worth protecting. In contrast, adversarial learning-based privacy protection has gained significant attention in non-FL contexts (Wang et al. 2023; Li et al. 2021, 2020). This strategy enables a more tailored approach by enhancing privacy only for sensitive attributes without affecting other useful information, thereby incurring minor accuracy compromise for the downstream target task.

Challenges. However, VFL poses several new challenges. First, clients hold distinct feature spaces and may inherently possess different sensitive attributes, which we refer to as *heterogeneous privacy*, such as race for image clients, political affiliation for text clients, and gender for audio clients. Second, enhancing local privacy inevitably degrades global target task accuracy, reflecting the “no free lunch” principle and leading to an *accuracy-privacy trade-off*. Third, these challenges must be addressed in a distributed setting, where clients perform local optimization without a global view, making it difficult to balance the trade-off, as privacy is adjusted locally while accuracy is achieved globally.

Solutions. We propose an equilibrium-driven VFL framework with selective privacy protection for sensitive attributes to enhance privacy with minor accuracy compromise. To facilitate efficient distributed optimization, we decompose the global objective, which captures the accuracy-privacy trade-off, into multiple local objectives tailored to each client. Since each client can only optimize its local model with conflicting objectives, the problem naturally lends itself to a game-theoretic interpretation. We introduce two key innovations: (1) a *NashCoder*, which incorporates a surrogate head to jointly optimize for accuracy and privacy; and (2) a Shapley-based adaptive decomposition strategy that dynamically reshapes local objectives in a non-uniform yet fair manner, thereby constructing a better equilibrium model.

Contributions. (1) We formulate a novel and practical VFL problem that accounts for heterogeneous privacy and accuracy-privacy trade-off. (2) We design *NashCoder* to selectively obfuscate sensitive attributes, enhancing local privacy with minor global accuracy compromise. (3) We propose a Shapley-based adaptive decomposition strategy that dynamically reshapes local objectives to construct a better equilibrium model in a distributed setting. (4) We provide theoretical analysis and evaluate our framework on three public datasets against five baselines, demonstrating a better accuracy-privacy trade-off across various settings. Extensive experimental results support our theoretical analysis.

2 Related Work

There are two lines of VFL: one assigns an active party (label holder) and passive parties that transmit partial gradients, while the other, adopted in this work and widely used in recent VFL studies, transmits embeddings to enable flexible

feature space partitioning and support arbitrary server models for feature fusion (Peng, Lu, and Xu 2024). Most existing VFL studies focus on two problems: (1) How to improve efficiency? Methods include compression techniques such as sparsification and quantization (Castiglia et al. 2022), asynchronous coordination (Castiglia, Wang, and Patterson 2023), and feature selection (Castiglia et al. 2023). (2) How to improve model effectiveness? VFL approaches include self-supervised (He et al. 2024), semi-supervised (Sun et al. 2023) and knowledge-distillation-based (Gao et al. 2024).

However, these works either rely on the flawed assumption that embeddings are privacy-preserving (which is not true) or adopt common privacy-enhancing strategies such as DP (Tran et al. 2023), HE (Gong et al. 2023), and MPC (Huang et al. 2023), some of which target attribute inference attacks (Luo et al. 2021). Such one-size-fits-all strategies often degrade the performance of downstream target task and incur excessive privacy overhead.

Adversarial learning-based methods (Wang et al. 2023; Li et al. 2021, 2020) share a similar privacy-enhancing motivation to obfuscate sensitive attributes. However, they are limited to non-FL settings and are not applicable to VFL setting, where privacy is adjusted locally while accuracy is achieved globally. Put differently, in VFL, the conflicting objectives among clients make distributed optimization challenging, as each client must balance global accuracy gains with local privacy costs without a global view.

3 Problem Formulation

To clearly formulate our problem, we clarify the usage of several symbols: $[\cdot]$ denotes a set of elements, $\circ$ represents the composition of functions, and $\{\cdot\}$ denotes sets of features, labels, or models depending on the context.

We consider a VFL system consisting of K clients and a server. The dataset $\mathbf{x} \in \mathbb{R}^{N \times M}$ is vertically partitioned among the K clients, where N is the number of samples and M is the feature dimension. We define the dataset for client k ($k \in [K]$) as $\mathbf{x}_k \in \mathbb{R}^{N \times M_k}$ ($M = \sum_{k=1}^K M_k$). The i -th row of $\mathbf{x}$ corresponds to a sample x^i ($i \in [N]$), where each sample x^i is composed of feature subsets held by each client k , denoted as x_k^i , such that $x^i = \{x_1^i, \dots, x_K^i\}$. Each x^i is associated with a target task label y^i and privacy task labels $\{z_k^i\}_{k=1}^K$, each corresponding to client-specific sensitive attributes. For example, if the global target task is emotion recognition, sensitive attributes may vary across different clients: race for images, political affiliation for text, and gender for audio. These attributes are unrelated to the global target task and are not in the form of easily isolated features, thus posing privacy risks (e.g., vulnerability to attribute inference attacks). Each client k locally holds an encoder model, parameterized by θ_k , implementing an embedding function $h_k(\cdot)$. The server holds the server head model, parameterized by θ_0 , and a loss function $\ell(\cdot)$ for the target task, which combines embeddings received from clients.

Threat Model. A potential external attacker (or curious server) may gain access to the embeddings and maliciously infer sensitive attributes (e.g., gender or race) unrelated to the target task, as illustrated in Fig. 1(b). This is known as

an attribute inference attack, which can be formulated as a classification problem where the attacker feeds the embeddings into an adversarial classifier for prediction. However, in practice, clients lack knowledge of the attacker’s model and only know which attributes (e.g., gender) are sensitive to them. To mimic the potential attacker’s behavior, each client k locally trains a surrogate head g_k to approximate the attribute inference attack and employs a privacy loss function $\ell_k(\cdot)$ to measure the inference ability, as part of our approach detailed in Section 4.2. A similar threat model has been adopted in several privacy-related works in non-FL contexts (Li et al. 2020, 2021; Wang et al. 2023).

Unlike standard VFL formulations, we explicitly account for heterogeneous privacy and accuracy-privacy trade-off, both posing key challenges in real-world distributed settings, and integrate them into the global VFL objective:

$$\max_{\Theta} F(\Theta) := u(\Theta) - \alpha \cdot \sum_{k \in [K]} c_k(\theta_k), \quad (1)$$

where $\Theta = \{\theta_0, \theta_1, \dots, \theta_K\}$ represents the global model, α controls the trade-off between accuracy and privacy, $u(\cdot)$ is a predefined accuracy utility function that quantifies the global model’s performance on the target task, and $c_k(\cdot)$ is a predefined privacy cost function that captures the potential privacy risk for client k , with details deferred to Section 4.2. Our objective is to train the global model Θ in a distributed manner that maximizes the global value function $F(\Theta)$, defined as the difference between the globally achieved accuracy utility and the sum of local privacy costs.

While the global objective is well-defined and practical in real-world scenarios, solving it is challenging due to the distributed nature of feature-partitioned data. To enable distributed training, we decompose the global value function into a set of local value functions, one for each client k :

$$f_k(\Theta) := q_k \cdot u(\Theta) - \alpha \cdot c_k(\theta_k), \quad \forall k \in [K], \quad (2)$$

where q_k is a decomposition weight that satisfies $\sum_k q_k = 1$. This decomposition allows us to express the global value function as $F(\Theta) := \sum_{k \in [K]} f_k(\Theta)$, enabling each client to focus on maximizing its local value, i.e., $\max_{\Theta} f_k(\Theta)$. Let $\mathbf{q} = \{q_1, \dots, q_K\}$ denote the decomposition vector. The choice of decomposition for $\mathbf{q}$ can significantly influence the final training outcome, highlighting the importance of an effective decomposition strategy, detailed in Section 4.3.

A Novel Equilibrium Perspective. Although the global value function $F(\Theta)$ in Eq. (1) can be decomposed into multiple local value functions $f_k(\Theta)$ in Eq. (2), each client k cannot directly maximize its local value $f_k(\Theta)$ since it only maintains its local model θ_k , rather than the entire model Θ , i.e., the absence of a global view. To explicitly capture this dependency, we express the local value function $f_k(\Theta)$ as $f_k(\theta_k, \Theta_{-k})$, where $\Theta_{-k} = \{\theta_{k'}\}_{k' \neq k}$ denotes the rest of the model excluding θ_k . This lack of global control naturally leads to a *game-theoretic interpretation*, in which each client k is a *player* aiming to maximize its local value $f_k(\theta_k, \Theta_{-k})$ by optimizing its local model θ_k under Θ_{-k} . In this multi-agent setting, a *Nash Equilibrium (NE)* provides a natural solution concept, representing a *stable state* in which no client can improve its local value $f_k(\theta_k, \Theta_{-k})$ by unilaterally adjusting its local model θ_k .

Definition 3.1 (Equilibrium Model). A global model Θ^{NE} is an equilibrium model if, for all $k \in [K]$,

$$f_k(\theta_k^{\text{NE}}, \Theta_{-k}^{\text{NE}}) \geq f_k(\theta'_k, \Theta_{-k}^{\text{NE}}), \quad \forall \theta'_k \neq \theta_k^{\text{NE}}. \quad (3)$$

Here, we focus on the equilibrium conditions among the K clients, excluding the server. Since all clients adjust their local models based on the same server head θ_0 , i.e., their objectives remain aligned with θ_0 , we can focus on optimizing the local models $\{\theta_1, \dots, \theta_K\}$ in a distributed manner.

4 Methodology

4.1 Overview of our VFL Framework

To illustrate the core idea of VFL, we first describe an idealized training algorithm, disregarding practical communication and computation constraints. The training proceeds in R global rounds, where each round r consists of local updates performed independently by each client.

Idealized Training. At the start of round r , each client k initializes its local model as $\theta_k^{r, t_0} = \theta_k^{r-1}$, using the converged local model from the previous round, and trains it using local dataset $\mathbf{x}_k$. The local training process involves iteratively applying (stochastic) gradient descent, updating $\theta_k^{r, t}$ at each local step t , and continuing until convergence. Since each client k requires embeddings from other clients $\{k'\}_{k' \neq k}$ and server head θ_0 to compute its local partial gradients, information exchange is essential. However, communication occurs only at the end of each round r . Consequently, during local updates in round r , client k only has access to the embeddings from other clients $\{k'\}_{k' \neq k}$ from the last round. Let $h_k(\theta_k^r; \mathbf{x}_k)$ denote the embeddings generated by client k using its locally converged model θ_k^r . We define $\Phi^r = \{h_k(\theta_k^r; \mathbf{x}_k)\}_{k=1}^K$ as the concatenated embeddings from all clients at the end of round r . During local updates in round r , client k updates its local encoder θ_k based on *fresh embeddings* $h_k(\theta_k^{r, t}; \mathbf{x}_k)$ computed using its current encoder $\theta_k^{r, t}$, and *stale embeddings* from other clients $\{k'\}_{k' \neq k}$ from the last round, denoted as $\tilde{\Phi}_{-k}^r = \{h_{k'}(\theta_{k'}^{r-1}; \mathbf{x}_{k'})\}_{k' \neq k}$. At the end of each round r , clients upload their fresh embeddings to the server, which then redistributes the concatenated embeddings Φ^r and server head θ_0^r to all clients for the next round of training. Equivalently, by the end of each round r , each client k updates its local model θ_k^r by solving $\theta_k^r = \arg \max_{\theta_k} f_k(\theta_k, \Theta_{-k}^{r-1})$, while the global model Θ^r is updated by composing the locally updated models $\{\theta_k^r\}$.

Practical Training. While conceptually clear, the above idealized training paradigm poses significant computational and communication challenges. Each global round can be time-consuming, as it may require many local updates for model convergence. Besides, sharing embeddings over the entire dataset causes substantial communication overhead, which is also impractical. To mitigate computational and communication overhead, we adopt two practical training strategies: (1) Fixed number of local updates: We perform Q local updates per round to improve efficiency while maintaining effective model training. (2) Mini-batch training: Instead of updating on the entire dataset, training is performed

on a sampled mini-batch $\mathcal{B}$. These two efficiency improvement strategies have been widely adopted in recent VFL works (Castiglia et al. 2022, 2023; Castiglia, Wang, and Paterson 2023) and are orthogonal to our work, which focuses on heterogeneous privacy and accuracy-privacy trade-off.

4.2 NashCoder: Selective Privacy Protection

We now present *NashCoder*, which replaces the standard encoder in VFL and leverages adversarial training to selectively obfuscate (client-specific) sensitive attributes that are otherwise difficult to isolate and eliminate. We illustrate the proposed training methodology in Fig. 2. The training of *NashCoder* is guided by the following two conflicting objectives, expressed from an information perspective: (1) *Global accuracy objective*: The output embedding h_k should capture as much useful information as possible to collaborate with other clients and achieve high accuracy on the global target task. (2) *Local privacy objective*: The output embedding h_k should convey as little sensitive attribute information as possible to enhance local privacy. For the privacy objective, as discussed in the threat model in Section 3, since the client lacks knowledge of the attacker’s model and only knows which attribute is sensitive to itself, each client k locally trains a surrogate head g_k to approximate the behavior of a potential attacker with the following objective:

$$\min_{g_k} \frac{1}{N} \sum_{i=1}^N \ell_k(g_k \circ h_k(\theta_k; x_k^i), z_k^i), \quad (4)$$

where $\ell_k(\cdot)$ is the local privacy task loss measuring the surrogate head’s ability to infer sensitive attribute. To jointly optimize global accuracy and local privacy in a distributed setting, the client locally trains the *NashCoder* θ_k and the surrogate head g_k under a given q_k and α by solving the following problem: $\min_{\theta_k} [q_k \cdot \ell(\cdot) - \alpha \cdot \min_{g_k} \ell_k(\cdot)]$, where $\ell(\cdot)$ represents the global target task loss.

Furthermore, in practice, the globally achieved accuracy provides utility to all participants, modeled by a server-defined utility function $u(\cdot)$, which quantifies the real-world utility (e.g., monetary benefits) of the global model’s performance. Similarly, local privacy leakage imposes a local cost on each client k , modeled by a client-defined cost function $c_k(\cdot)$, which captures real-world costs (e.g., monetary loss) as a function of the privacy attack’s effectiveness. For training purposes, we let $u(\cdot)$ depend on the target task loss instead of the accuracy, and $c_k(\cdot)$ depend on the local privacy task loss instead of the attack accuracy. By combining the local privacy cost and the decomposed global accuracy utility, each client solves the following min-max optimization problem: $\min_{\theta_k} [\alpha \cdot \max_{g_k} c_k(g_k, \theta_k) - q_k \cdot u(\theta_k, \Theta_{-k})]$, and local updates alternate between θ_k and g_k in practice. Thus, given a q_k (e.g., $q_k = \frac{1}{K}$, $\forall k$) and α , we train θ_k using the following objective to address heterogeneous privacy and balance trade-off between global accuracy and local privacy:

$$\min_{\theta_k} \frac{1}{N} \sum_{i=1}^N \left[\underbrace{\alpha \cdot c_k(\ell_k(g_k \circ h_k(\theta_k; x_k^i); z_k^i))}_{\text{Local privacy cost}} - \underbrace{q_k \cdot u(\ell(\tilde{\theta}_0 \circ \{h_k(\theta_k; x_k^i), \tilde{\Phi}_{-k}\}; y^i))}_{\text{Decomposed global accuracy utility}} \right]. \quad (5)$$

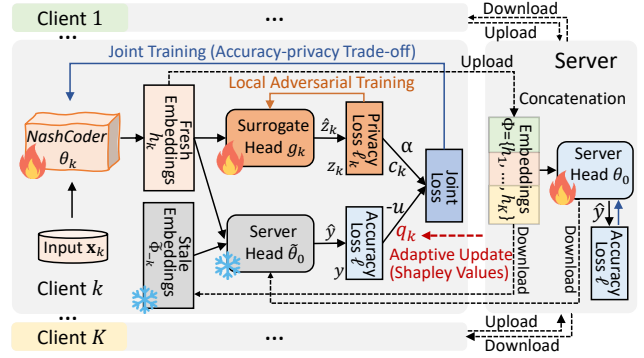


Figure 2: (1) Each client k selectively obfuscates client-specific sensitive attributes via *NashCoder* θ_k by incorporating a surrogate head g_k for local adversarial training. (2) Server adaptively updates the decomposition weight q_k for each client k . Downloads occur at the start of each round $r^{t=0}$, and uploads occur at the end $r^{t=Q}$. The spark denotes training, while the snowflake denotes a fixed state.

Note that $u(\cdot)$ and $c_k(\cdot)$ are predefined by the problem and not designed by our algorithm; their role is simply to convert different losses into meaningful values for joint accuracy and privacy training. As a toy example for illustration, we may set $u(x) = -x$, where the globally achieved accuracy utility is the negative of the target task loss; similarly for c_k . Besides, $\tilde{\theta}_0$ and $\tilde{\Phi}_{-k}$ are only shared at the start of each round for efficiency, which is widely adopted by VFL paradigms (Castiglia et al. 2022, 2023). Such efficiency improvements are orthogonal to our main contribution prioritizing accuracy-privacy trade-off.

4.3 Adaptive Decomposition

So far, we have assumed a fixed decomposition vector $\mathbf{q}$ and a given q_k for each client k in Eq. (5). However, the choice of $\mathbf{q}$ clearly impacts training outcomes, and it remains unclear which decomposition should be used. The simplest strategy is a uniform decomposition, i.e., $q_k = \frac{1}{K}$ for each client k , which promotes equal contributions to the global model performance, as shown in Fig. 3 (left). However, this method is clearly suboptimal, as different clients contribute unequally to the global target task depending on the importance of their provided embeddings. Another decomposition strategy that naturally arises is the weight-based approach, but it is still essentially a fixed decomposition, and it remains unclear how to fairly assign different q_k values to each client k in practice. Notably, accuracy is achieved globally and privacy is adjusted locally in VFL; thus, the trade-off balance differs from the non-FL setting and naturally lends itself to a game-theoretic interpretation. Different decomposition strategies for $\mathbf{q}$ lead to different games and result in different equilibrium models. Our goal is to find a better decomposition strategy to construct a better equilibrium model by reshaping the value functions $\{f_k\}_{k \in [K]}$ in Eq. (2), thereby enhancing overall outcomes reflected in a better trade-off.

To this end, we propose an adaptive decomposition strategy from an equilibrium perspective, which dynamically ad-

justs the decomposition weights $\{q_k\}_{k \in [K]}$ based on each client's contribution. Specifically, we use the *Shapley value* to evaluate each client's marginal contribution to the global target task at the end of each round in a non-uniform yet fair manner, which guides the decomposition of $\mathbf{q}$ in the next round. For clarity, we omit $\ell(\cdot)$ and other unnecessary notations below. To ensure $\sum_{k=1}^K q_k = 1$, we use a normalized utility for any subset of clients $S \subseteq [K]$, computed as: $\hat{u}(\Theta^r[S]) = \frac{u(\Theta^r[S]) - u(\Theta^r[\emptyset])}{u(\Theta^r[S]) - u(\Theta^r[\emptyset])}$. Here, $u(\Theta^r[S])$ is the accuracy utility achieved by coalition S , and $u(\Theta^r[\emptyset])$ is the utility achieved by the empty coalition. The marginal contribution of each client k to the global model performance is quantified by how much the accuracy utility increases when adding k to a coalition S . For any coalition S , the marginal contribution of client k is $\hat{u}(\Theta^r[S \cup \{k\}]) - \hat{u}(\Theta^r[S])$. Thus, the decomposition weight q_k is computed by: $q_k^{r+1} = \sum_{S \subseteq [K] \setminus \{k\}} \frac{|S|!(K-|S|-1)!}{K!} \cdot (\hat{u}(\Theta^r[S \cup \{k\}]) - \hat{u}(\Theta^r[S]))$.

An important motivation for using the Shapley value is that the decomposition should be entirely determined by the server, based solely on each client's marginal contribution rather than client-reported local values f_k that incorporate client-defined privacy costs c_k ; that is, the decomposition is privacy-cost-agnostic to avoid imbalances caused by c_k . Moreover, most existing studies (Liu et al. 2024, 2022b; Castiglia et al. 2022) suggest that the number of clients K is typically small in VFL, as the number of splittable features (modalities) is limited in practice (unlike HFL, which involves many clients). Thus, the computational cost of calculating exact q_k on a powerful server is relatively minor. We offer a sampling-based approximation for q_k to address potential scalability issues in extreme cases where K is large. However, we do not claim this efficiency improvement as our core contribution, as it is orthogonal to our main objective of balancing the trade-off from an equilibrium view.

4.4 Theoretical Analysis

We first demonstrate that, under common smoothness assumptions, our idealized distributed training algorithm with adaptive decomposition converges to an equilibrium model.

Theorem 4.1 (Convergence). Define $C \triangleq \sum_k \sum_{S \subseteq [K] \setminus \{k\}} \frac{|S|!(K-|S|-1)!}{K!}$. If L_1, L_2, L_3 satisfy $2L_1L_3C + L_2 < 1$, then the global model Θ and decomposition vector $\mathbf{q}$ obtained by our distributed algorithm converge to stable $\Theta^\dagger$ and $\mathbf{q}^\dagger$. Moreover, $\Theta^\dagger$ is an equilibrium model under $\mathbf{q}^\dagger$.

Furthermore, given any decomposition vector $\mathbf{q}$ and trade-off parameter α , we bound the gap between the corresponding NE and the global optimum under common smoothness assumptions parameterized by constants L, L_0, L_c , and L_q , where $L_c = \max_{k \in [K]} L_{c,k}$ and $L_q = \max_{k \in [K]} L_{q,k}$, which depend on the privacy cost functions $\{c_k\}_{k \in [K]}$ and the decomposition weights $\{q_k\}_{k \in [K]}$, respectively.

Theorem 4.2 (Bound between NE and the global optimum). If $1 - L_q(K-1) > 0$, let Θ^{OPT} denote the global optimum and $\Theta^{NE}(\mathbf{q})$ denote the NE solution of $F(\Theta)$ under $\mathbf{q}$. Then, the gap between $F(\Theta^{OPT})$ and $F(\Theta^{NE}(\mathbf{q}))$ is upper bounded by: $\|F(\Theta^{OPT}) - F(\Theta^{NE}(\mathbf{q}))\| \leq \alpha L(L_0 + \frac{L_c K}{1 - L_q(K-1)})$.

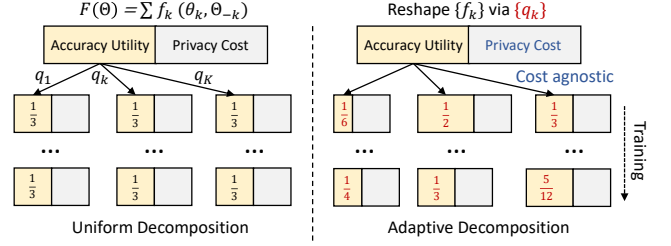


Figure 3: We aim to construct a better equilibrium model by adaptively reshaping value functions via $\mathbf{q}$ decomposition.

This bound aligns with the intuition that the NE solution approaches the global optimum as $\alpha \rightarrow 0$ (i.e., all clients care only about global accuracy utility and ignore local privacy cost, as in standard VFL). Detailed assumptions and proofs of Theorems 4.1 and 4.2 are provided in the Appendix (Peng 2025). “Theory vs. Practice” discussion is deferred to Section 5.4, showing that experimental results under various settings align with our theoretical analysis.

5 Experiments

We aim to investigate the following three questions in our experiments to validate our framework and theoretical analysis. The corresponding results and analyses are presented in Sections 5.2, 5.3, and 5.4, respectively: **Q1:** Can we achieve a better trade-off when sensitive attributes are homogeneous across clients? **Q2:** Can we achieve a better trade-off when sensitive attributes are heterogeneous? **Q3:** How and why do key components and parameters affect performance?

5.1 Experimental Setup

Datasets: We adopt three publicly available datasets covering different VFL applications and tasks. (1) *IEMOCAP* consists of visual, text, and audio modalities recorded from multiple actors during conversations (Busso et al. 2008). The target task is set as emotion recognition, while the privacy task is set as gender prediction, as gender is unrelated to the target task but can be maliciously inferred. (2) *MIMIC-III* is a collection of medical records of patients admitted to ICUs. Each sample includes 76 features, such as vital signs and medications within a period (Johnson et al. 2016). The target task is set as the prediction of in-hospital mortality (IHM). The privacy task is set as predicting a length of stay (LOS) longer than two days, as this attribute can be maliciously inferred for strategic insurance pricing. (3) *CelebA* contains a large-scale collection of facial images, each labeled with 40 binary attributes (Liu et al. 2015). The predictions of “smiling” and “gray hair” are set as different global target tasks, while the predictions of “male” and “young” are set as different privacy tasks.

Baselines: (1) *Standard VFL* transmits embeddings instead of feature-partitioned data but is vulnerable to potential attacks. (2) *DP-Gaussian* applies DP in VFL by injecting Gaussian noise to enhance privacy (Wang et al. 2024). (3) *DP-Laplacian* adopts DP in VFL by injecting Laplacian noise (Ovi et al. 2023). (4) *C-VFL* employs a compression

strategy by pruning embeddings, which enhances privacy as a byproduct (Castiglia et al. 2022). (5) *Ours with uniform decomposition* ($q_k = \frac{1}{K}, \forall k$) serves as an important baseline to highlight the advantages of our adaptive decomposition.

Implementation and Metrics: We run all experiments on NVIDIA GeForce RTX 4090 GPUs. Essential implementation details and metrics are provided below. For some imbalanced datasets, we use the F1 score instead of accuracy. In the following, ($\uparrow$) and ($\downarrow$) indicate that higher or lower values correspond to better performance. For IEMOCAP, we partition the data across three modalities for three clients. The privacy task metric is the F1 score for gender prediction ($\downarrow$), and the target task metric is the weighted F1 score for emotion recognition ($\uparrow$). For MIMIC-III, we partition the data across 76 features for two clients, and increase the number of clients K for scalability experiments in Section 5.4. The privacy task metric is the F1 score for LOS >2 ($\downarrow$), and the target task metric is the F1 score for in-hospital mortality (IHM) ($\uparrow$), since only 16% of samples are positive (i.e., most patients did not experience IHM). For CelebA, each image is vertically split into two parts for two clients, which is commonly used for feature partitioning in VFL (Sun et al. 2023; Castiglia et al. 2022). We use the attacker’s accuracy for predicting “male” and “young” as metrics for (heterogeneous) privacy task performance ($\downarrow$), and accuracy for predicting “smiling” and “gray hair” as metrics for two different target tasks ($\uparrow$). Notably, the privacy metrics, evaluated from the attacker’s perspective, align with those used in adversarial-learning-based methods in non-FL contexts (Li et al. 2020, 2021; Wang et al. 2023), and are widely adopted in attribute inference attacks (Liu et al. 2022a).

5.2 Results of Homogeneous Sensitive Attributes

To answer **Q1**, we first set the sensitive attributes of different clients to be homogeneous (note that clients are unaware of this homogeneity and c_k may differ). For each dataset, we present two types of results. First, we compare our method against baselines to observe the *overall trend* in the accuracy–privacy trade-off. Each method (except for the standard VFL, which ignores privacy) is repeated five times under different α , i.e., each point (with the same marker) represents the performance of a method under a fixed α . The horizontal axis indicates the global target task performance collaboratively achieved by all clients, while the vertical axis indicates the average privacy performance across all clients. As shown in Fig. 4 on three datasets, our method consistently approaches the optimal accuracy-privacy trade-off (i.e., the lower right corner indicates better target task accuracy from the server’s view and worse privacy task accuracy from the surrogate’s view) compared to baselines, as it selectively obfuscates sensitive attributes with less accuracy compromise. Notably, our adaptive decomposition achieves better performance by assigning higher weights to clients with greater contributions in a non-uniform yet fair manner.

Second, we show each client’s *individual performance* under a fixed α . The horizontal axis still denotes the target task performance, while the vertical axis shows each client’s individual privacy task performance (solid, hollow, and bordered markers denote clients C1, C2, and C3). As shown in

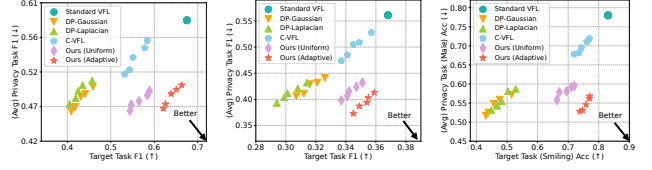


Figure 4: Overall accuracy-privacy trade-off trends on three datasets. From left to right: IEMOCAP, MIMIC-III, CelebA.

Fig. 5 on three datasets, our method achieves a better accuracy–privacy trade-off, as clients not only collaboratively improve global accuracy utility but also reduce local privacy costs to maximize their individual value. Put differently, we construct a better equilibrium model by adaptively reshaping the value functions, thereby improving overall outcomes.

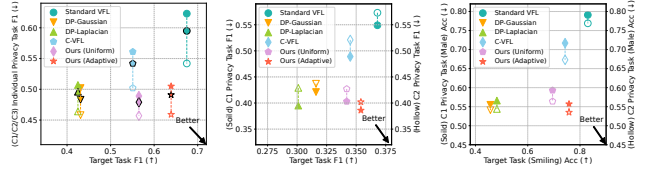


Figure 5: Client-level accuracy-privacy trade-offs on three datasets. From left to right: IEMOCAP, MIMIC-III, CelebA.

5.3 Results of Heterogeneous Sensitive Attributes

To answer **Q2**, we conduct experiments on CelebA by assigning heterogeneous privacy tasks (“male” and “young”) to two clients (each with half of the facial image) associated with the same target task. Besides, we perform experiments on two different target tasks (“smiling” and “gray hair”) to demonstrate broad applicability. As shown in Fig. 6, our method achieves a better accuracy-privacy trade-off than baselines. Thus, it is suitable for diverse personalized privacy objectives commonly seen in real-world scenarios, where accuracy is achieved globally and privacy is adjusted locally. Notably, the server adjusts decomposition weights solely based on each client’s utility contribution, regardless of the privacy cost reported by the client (i.e., cost-agnostic).

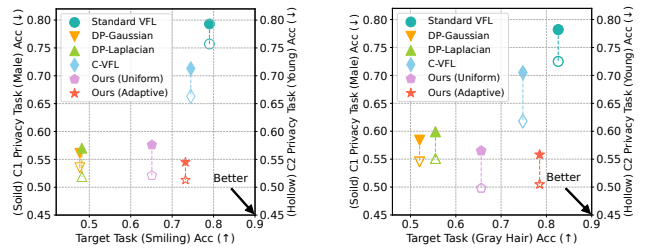


Figure 6: Trade-off in the heterogeneous privacy setting under different target tasks (left: smiling; right: gray hair).

5.4 Impact of Key Components and Parameters

To answer **Q3**, we conduct more experiments including ablation studies. Results on MIMIC-III are presented, with similar trends observed on the other two datasets.

Impact of Adversarial Training. As shown in Fig. 7 (a), our method selectively obfuscates sensitive attributes without affecting other useful information via adversarial training. Thus, for the target task, our method closely approaches the performance of standard VFL, which is impractical in the real world as it ignores privacy cost. Besides, although the surrogate head is a relatively shallow classifier and efficient to train in practice, we further investigate its additional computational cost. As shown in Fig. 7 (b), the smaller the shadow areas, the better, i.e., privacy can be enhanced via adversarial training. Notably, there is no additional computational cost during inference, which makes it practical for most VFL applications that prioritize privacy.

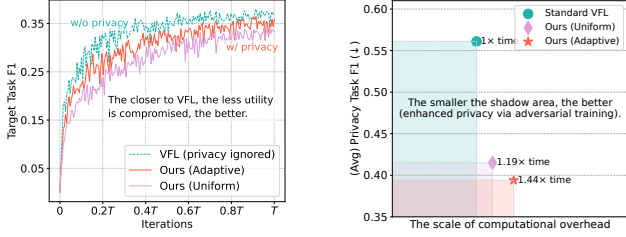


Figure 7: (a) Convergence (b) Impact of adversarial training.

Impact of Adaptive Decomposition. As shown in Fig. 8 (left), our method with adaptive decomposition achieves a better accuracy-privacy trade-off compared to uniform decomposition. This is because the server dynamically adjusts q_k to reshape the value functions f_k for each client k based on their past contributions in a non-uniform yet fair manner, thereby helping construct a better equilibrium model in a distributed manner and improving overall outcomes.

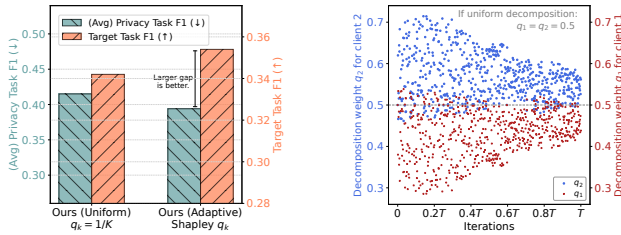


Figure 8: Impact of adaptive decomposition and dynamic q_k .

Impact of α . As shown in Fig. 9(a), increasing α causes the value function to assign greater weight to privacy cost. However, changes in α are applied equally to all clients, i.e., α can only balance the trade-off between accuracy and privacy, but it cannot reshape the value functions to construct a better equilibrium model as our framework aims to do. Put differently, changing α affects accuracy utility and privacy cost simultaneously, but cannot lead to higher overall value.

Impact of Q . As shown in Fig. 9(b), increasing Q reduces communication frequency under a fixed total number of updates, improving communication efficiency but degrading performance, as each client relies on stale information for more local updates and engages in fewer global rounds.

Impact of K . As shown in Fig. 9(c), increasing K slightly degrades the global target task performance, as a fixed number of feature dimensions is distributed across more clients,

causing each client to maintain a narrower feature space and rely on a larger proportion of stale information during local updates within each round. Moreover, a larger K also decreases the F1 score of the privacy task, as fewer (split) features make it harder for the surrogate model to infer sensitive attributes. Notably, most VFL papers (Liu et al. 2024; Castiglia et al. 2022; Liu et al. 2022b) note that K is typically small, as the number of splittable features (modalities) is limited in practice, making scalability a relatively minor concern compared to the accuracy-privacy trade-off in VFL.

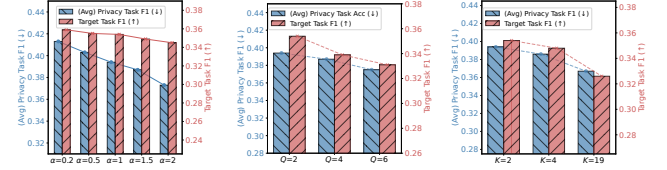


Figure 9: (a) Impact of α . (b) Impact of Q . (c) Impact of K .

Theory vs. Practice. Extensive experimental results in Figs. 4 to 6 under both homogeneous and heterogeneous sensitive-attribute settings, the convergence and improved privacy performance via local adversarial training in Fig. 7, the adaptive decomposition results in Fig. 8, and the evaluations across various trade-off parameters α , communication frequencies Q , and numbers of clients K in Fig. 9 collectively demonstrate the effectiveness of our framework.

Importantly, these results align with our theoretical analysis in Theorem 4.1, which shows that the global model Θ and decomposition vector q , obtained through our distributed algorithm, converge to stable values $\Theta^\dagger$ and $q^\dagger$, where $\Theta^\dagger$ is an equilibrium model under $q^\dagger$. Such an equilibrium model achieves a better trade-off between global accuracy and local privacy. Furthermore, we show in Theorem 4.2 that our NE solution approaches the global optimum as $\alpha \rightarrow 0$ (i.e., when all clients care only about global accuracy and ignore local privacy costs, as in standard VFL), thereby highlighting the practicality of our distributed algorithm in real-world scenarios, where clients not only benefit from global accuracy gains via collaboration but also incur local privacy costs that must be carefully balanced in a distributed setting.

6 Conclusion

In conclusion, we formulate a novel and practical VFL problem that explicitly accounts for heterogeneous privacy and the accuracy-privacy trade-off, and propose an equilibrium-driven VFL framework to solve it via global objective decomposition. We introduce two key innovations: (1) *Nash-Coder*, which incorporates a surrogate head to jointly optimize global accuracy and local privacy; and (2) an adaptive decomposition strategy based on Shapley values, which dynamically decomposes the global objective for distributed optimization from an equilibrium perspective. We theoretically analyze our framework and empirically evaluate it on three public datasets against five baselines, demonstrating significant improvements in the accuracy-privacy trade-off under various privacy settings. Extensive experimental results further support our theoretical analysis.

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7 Appendix

In the appendix, we provide supplementary details to support the main paper as follows:

- Appendix 7.1: Notations.
- Appendix 7.2: Proof of Theorem 4.1.
- Appendix 7.3: Proof of Theorem 4.2.
- Appendix 7.4: Sampling-based approximation of q_k to further reduce computational costs.
- Appendix 7.5: Implementation and reproducibility.
- Appendix 7.6: Preliminary experimental results on baselines.
- Appendix 7.7: Additional experimental results.
- Appendix 7.8: Additional discussion.

7.1 Notations

We provide a complete list of notations below for clarity.

Table 1: Notations

Notation	Definition
N	Number of data samples, $i \in [N]$.
M	Number of feature dimensions, $M = \sum_{k=1}^K M_k$.
K	Number of clients, as shown in Fig. 1(a), $k \in [K]$.
$\mathbf{x}$	Global dataset across all clients, $\mathbf{x} \in \mathbb{R}^{N \times M}$.
$\mathbf{x}_k$	Local (feature-partitioned) dataset for client k , $\mathbf{x}_k \in \mathbb{R}^{N \times M_k}$.
$\mathbf{y}$	Global target task labels (e.g., emotion).
$\mathbf{z}_k$	Sensitive attributes for client k (privacy labels, e.g., gender or race).
$\mathcal{B}$	Randomly sampled mini-batch of size B .
Θ	Global model, $\Theta = \{\theta_0, \theta_1, \dots, \theta_K\}$.
θ_0	Server head model ($k = 0$).
$\tilde{\theta}_0$	Stale server head model used by each client.
θ_k	Local (encoder) model for client k .
$h_k(\cdot)$	Embedding function for client k .
$\tilde{\Phi}_{-k}$	Stale embeddings used by client k .
g_k	Surrogate head for client k .
α	Trade-off parameter balancing accuracy utility and privacy cost.
$u(\cdot)$	Accuracy utility function, predefined by the server.
$c_k(\cdot)$	Privacy cost function, predefined by each client k .
Q	VFL communication frequency, i.e., every Q local updates.
R	Total number of global rounds, $r \in [R]$.
T	Total number of local updates, $t \in [T]$, $T = R \times Q$.
η	Learning rate.
$\mathbf{q}$	Decomposition vector, i.e., $\mathbf{q} = \{q_1, \dots, q_K\}$
q_k	Positive decomposition weight for client k .
$F(\Theta)$	Global objective (value function), $F(\Theta) := u(\Theta) - \alpha \cdot \sum_{k \in [K]} c_k(\theta_k)$.
$f_k(\Theta)$	Decomposed local objective for client k , $f_k(\Theta) := q_k \cdot u(\Theta) - \alpha \cdot c_k(\theta_k)$.
C	Constant for proof of Theorem 4.1.
L_1, L_2, L_3	L-smoothness constants for proof of Theorem 4.1.
L, L_0, L_c, L_q	L-smoothness constants for proof of Theorem 4.2.

7.2 Proof of Theorem 4.1

Assumption 1. There exists a positive constant $L_1 \leq \infty$ such that, for all Θ_1 and Θ_2 , the normalized accuracy utility satisfies: $|\hat{u}(\Theta_1) - \hat{u}(\Theta_2)| \leq L_1 \|\Theta_1 - \Theta_2\|$.

Assumption 2. Let $\Theta^*(\Theta^0, \mathbf{q}) = \arg \min_{\Theta} F(\Theta | \Theta^0, \mathbf{q})$ denote the optimal model for a given decomposition vector $\mathbf{q}$ and previous model Θ^0 . There exist positive constants L_2 and L_3 such that,

for all Θ_1^0, Θ_2^0 and $\mathbf{q}_1, \mathbf{q}_2$, the optimal solution satisfies:

$$\|\Theta^*(\Theta_1^0, \mathbf{q}) - \Theta^*(\Theta_2^0, \mathbf{q})\| \leq L_2 \|\Theta_1^0 - \Theta_2^0\|, \quad (1)$$

$$\|\Theta^*(\Theta^0, \mathbf{q}_1) - \Theta^*(\Theta^0, \mathbf{q}_2)\| \leq L_3 \|\mathbf{q}_1 - \mathbf{q}_2\|. \quad (2)$$

Theorem 4.1 (Convergence). Suppose that Assumptions 1 and 2 hold and define the constant $C \triangleq \sum_{k \in [K]} \sum_{S \subseteq [K] \setminus \{k\}} \frac{|S|!(K-|S|-1)!}{K!}$. If L_1, L_2 , and L_3 satisfy $2L_1L_3C + L_2 < 1$, then the global model Θ and the decomposition vector $\mathbf{q}$ obtained by our distributed algorithm converge to stable values $\Theta^\dagger$ and $\mathbf{q}^\dagger$. Moreover, $\Theta^\dagger$ is an equilibrium model under $\mathbf{q}^\dagger$.

Proof. Consider the decomposition vectors $\mathbf{q}^{r+1}$ and $\mathbf{q}^r$ determined by our adaptive decomposition strategy in two consecutive rounds. By the definition of q_k , we have

$$|q_k^{r+1} - q_k^r| \quad (3)$$

$$\stackrel{(i)}{\leq} \left(\sum_{S \subseteq [K] \setminus \{k\}} \frac{|S|!(K-|S|-1)!}{K!} \right) |\hat{u}(\Theta^r[S \cup \{k\}]) - \hat{u}(\Theta^r[S]) - (\hat{u}(\Theta^{r-1}[S \cup \{k\}]) - \hat{u}(\Theta^{r-1}[S]))| \quad (4)$$

$$\stackrel{(ii)}{\leq} \sum_{S \subseteq [K] \setminus \{k\}} \frac{|S|!(K-|S|-1)!}{K!} (|\hat{u}(\Theta^r[S \cup \{k\}]) - \hat{u}(\Theta^{r-1}[S \cup \{k\}])| + |\hat{u}(\Theta^r[S]) - \hat{u}(\Theta^{r-1}[S])|) \quad (5)$$

$$\stackrel{(iii)}{\leq} 2L_1 \sum_{S \subseteq [K] \setminus \{k\}} \frac{|S|!(K-|S|-1)!}{K!} \|\Theta^r - \Theta^{r-1}\|, \quad (6)$$

where (i) follows from the definition of q_k , (ii) follows from the triangle inequality, and (iii) follows from Assumption 1. Summing over k , we have

$$\|\mathbf{q}^{r+1} - \mathbf{q}^r\| \leq \sum_{k \in [K]} |q_k^{r+1} - q_k^r| \quad (7)$$

$$\stackrel{(i)}{\leq} 2L_1 \sum_{k \in [K]} \sum_{S \subseteq [K] \setminus \{k\}} \frac{|S|!(K-|S|-1)!}{K!} \|\Theta^r - \Theta^{r-1}\| \quad (8)$$

$$\stackrel{(ii)}{\leq} 2L_1C \|\Theta^r - \Theta^{r-1}\|, \quad (9)$$

where (i) follows from inequality (6), and (ii) follows from the definition of the constant $C \triangleq \sum_{k \in [K]} \sum_{S \subseteq [K] \setminus \{k\}} \frac{|S|!(K-|S|-1)!}{K!}$. Given the bound on $\|\mathbf{q}^{r+1} - \mathbf{q}^r\|$ in inequality (9), we can further bound $\|\Theta^{r+1} - \Theta^r\|$ as follows. According to our distributed training algorithm with the adaptive decomposition strategy, the models obtained after two consecutive training rounds satisfy:

$$\|\Theta^{r+1} - \Theta^r\| = \|\Theta^*(\Theta^r, \mathbf{q}^{r+1}) - \Theta^*(\Theta^{r-1}, \mathbf{q}^r)\| \quad (10)$$

$$= \|\Theta^*(\Theta^r, \mathbf{q}^{r+1}) - \Theta^*(\Theta^r, \mathbf{q}^r) + \Theta^*(\Theta^r, \mathbf{q}^r) - \Theta^*(\Theta^{r-1}, \mathbf{q}^r)\| \quad (11)$$

$$\stackrel{(i)}{\leq} \|\Theta^*(\Theta^r, \mathbf{q}^{r+1}) - \Theta^*(\Theta^r, \mathbf{q}^r)\| + \|\Theta^*(\Theta^r, \mathbf{q}^r) - \Theta^*(\Theta^{r-1}, \mathbf{q}^r)\| \quad (12)$$

$$\stackrel{(ii)}{\leq} L_3 \|\mathbf{q}^{r+1} - \mathbf{q}^r\| + L_2 \|\Theta^r - \Theta^{r-1}\| \quad (13)$$

$$\stackrel{(iii)}{\leq} (2L_1L_3C + L_2) \|\Theta^r - \Theta^{r-1}\| \quad (14)$$

$$\stackrel{(iv)}{\leq} (2L_1L_3C + L_2)^r \|\Theta^1 - \Theta^0\|, \quad (15)$$

where (i) follows from the inequality $\|a - c\| \leq \|a - b\| + \|b - c\|$, (ii) follows from Assumption 2, (iii) follows from inequality (9), and (iv) follows by recursively applying the inequality.

Since $2L_1L_3C + L_2 < 1$, it follows that Θ^r converges to some $\Theta^\dagger$, i.e., $\lim_{r \rightarrow \infty} \Theta^r = \Theta^\dagger$. Consequently, the decomposition vector $\mathbf{q}$ converges as well: $\lim_{r \rightarrow \infty} \mathbf{q}^r = \mathbf{q}^\dagger$. Finally, $\Theta^\dagger$ is a NE under $\mathbf{q}^\dagger$ since under convergence, $\Theta^\dagger$ is the best response to itself, i.e., $\Theta^\dagger = \Theta^*(\Theta^\dagger, \mathbf{q}^\dagger)$. $\square$

Theory vs. Practice. Extensive experimental results, including those in Figs. 4 to 6 under homogeneous and heterogeneous sensitive attributes, the convergence results and improved privacy performance via adversarial training in Fig. 7, the adaptive decomposition results in Fig. 8, and the evaluation across different trade-off parameters α , communication frequencies Q , and numbers of clients K (scalability) in Fig. 9, collectively demonstrate the effectiveness of our framework. Importantly, these results align with our theoretical analysis in Theorem 4.1, which shows that the global model Θ and decomposition vector $\mathbf{q}$, obtained through our distributed algorithm, converge to stable values $\Theta^\dagger$ and $\mathbf{q}^\dagger$, where $\Theta^\dagger$ is an equilibrium model under $\mathbf{q}^\dagger$. Such an equilibrium model achieves a better trade-off between global accuracy and local privacy.

7.3 Proof of Theorem 4.2

We define the global optimal solution as $\Theta^{\text{OPT}} = \arg \max_{\Theta} [u(\Theta) - \alpha \cdot \sum_k c_k(\theta_k)]$, and define the NE solution for client k as $\theta_k^{\text{NE}} = \arg \max_{\theta_k} [q_k \cdot u(\theta_k, \Theta_{-k}^{\text{NE}}) - \alpha \cdot c_k(\theta_k)]$. We also define a parameterized best-response function:

$$v_k(\theta_k \mid \Theta_{-k}, \alpha) = q_k \cdot u(\theta_k, \Theta_{-k}) - \alpha \cdot c_k(\theta_k), \quad (16)$$

where q_k , α , and Θ_{-k} are fixed for client k , and the optimization is over θ_k . The best-response strategy of client k is given by:

$$\theta_k^*(\Theta_{-k}, \alpha) = \arg \max_{\theta_k} v_k(\theta_k \mid \Theta_{-k}, \alpha). \quad (17)$$

Assumption 3. The global value function $F(\Theta)$ is L -Lipschitz continuous: $\|F(\Theta_1) - F(\Theta_2)\| \leq L \cdot \|\Theta_1 - \Theta_2\|$.

Assumption 4. For each client k , the best response $\theta_k^*(\Theta_{-k}, \alpha)$ satisfies the following conditions with respect to changes in α and Θ_{-k} :

$$\|\theta_k^*(\Theta_{-k}, \alpha_1) - \theta_k^*(\Theta_{-k}, \alpha_2)\| \leq L_c \cdot |\alpha_1 - \alpha_2|, \quad (18)$$

$$\|\theta_k^*(\Theta'_{-k}, \alpha) - \theta_k^*(\Theta''_{-k}, \alpha)\| \leq L_q \cdot \|\Theta'_{-k} - \Theta''_{-k}\|, \quad (19)$$

where $L_c = \max_{k \in [K]} L_{c,k}$, and $L_{c,k}$ captures how changes in α affect client k 's response under its cost function c_k . Similarly, $L_q = \max_{k \in [K]} L_{q,k}$, and $L_{q,k}$ captures how changes in Θ_{-k} affect client k 's response under its decomposition weight q_k .

Assumption 5. Let $\Theta^\dagger(\alpha) = \arg \max_{\Theta} [u(\Theta) - \alpha \sum_k c_k(\theta_k)]$. Then, $\Theta^\dagger(\alpha)$ satisfies the following condition with respect to changes in α : $\|\Theta^\dagger(\alpha_1) - \Theta^\dagger(\alpha_2)\| \leq L_0 \cdot |\alpha_1 - \alpha_2|$.

Theorem 4.2 (Bound between NE and the global optimum). Suppose that Assumptions 3-5 hold and $1 - L_q(K - 1) > 0$. Let Θ^{OPT} denote the global optimum and $\Theta^{\text{NE}}(\mathbf{q})$ denote the NE solution of $F(\Theta)$ under $\mathbf{q}$. Then, the gap between $F(\Theta^{\text{OPT}})$ and $F(\Theta^{\text{NE}}(\mathbf{q}))$ is upper bounded by:

$$\|F(\Theta^{\text{OPT}}) - F(\Theta^{\text{NE}}(\mathbf{q}))\| \leq \alpha L \left(L_0 + \frac{L_c K}{1 - L_q(K - 1)} \right). \quad (20)$$

Proof. We first define the global optimum that considers only the accuracy utility (i.e., ignoring the privacy cost) as:

$$\tilde{\Theta}^{\text{OPT}} = \arg \max_{\Theta} u(\Theta). \quad (21)$$

Then for each client k , we have:

$$\tilde{\theta}_k^{\text{OPT}} = \arg \max_{\theta_k} g_k(\theta_k \mid \tilde{\Theta}_{-k}^{\text{OPT}}, 0) = \theta_k^*(\tilde{\Theta}_{-k}^{\text{OPT}}, 0), \quad (22)$$

$$\theta_k^{\text{NE}} = \arg \max_{\theta_k} g_k(\theta_k \mid \tilde{\Theta}_{-k}^{\text{NE}}, \alpha) = \theta_k^*(\Theta_{-k}^{\text{NE}}, \alpha). \quad (23)$$

We can bound:

$$\|\tilde{\theta}_k^{\text{OPT}} - \theta_k^{\text{NE}}\| = \|\theta_k^*(\tilde{\Theta}_{-k}^{\text{OPT}}, 0) - \theta_k^*(\Theta_{-k}^{\text{NE}}, \alpha)\| \quad (24)$$

$$= \|\theta_k^*(\tilde{\Theta}_{-k}^{\text{OPT}}, 0) - \theta_k^*(\Theta_{-k}^{\text{NE}}, 0) + \theta_k^*(\Theta_{-k}^{\text{NE}}, 0) - \theta_k^*(\Theta_{-k}^{\text{NE}}, \alpha)\| \quad (25)$$

$$\stackrel{(i)}{\leq} \|\theta_k^*(\tilde{\Theta}_{-k}^{\text{OPT}}, 0) - \theta_k^*(\Theta_{-k}^{\text{NE}}, 0)\| + \|\theta_k^*(\Theta_{-k}^{\text{NE}}, 0) - \theta_k^*(\Theta_{-k}^{\text{NE}}, \alpha)\| \quad (26)$$

$$\stackrel{(ii)}{\leq} \left\| \theta_k^*(\tilde{\Theta}_{-k}^{\text{OPT}}, 0) - \theta_k^*(\Theta_{-k}^{\text{NE}}, 0) \right\| + L_c \alpha \quad (27)$$

$$\stackrel{(iii)}{\leq} L_q \left\| \tilde{\Theta}_{-k}^{\text{OPT}} - \Theta_{-k}^{\text{NE}} \right\| + L_c \alpha \quad (28)$$

$$\stackrel{(iv)}{\leq} L_q \sum_{k' \neq k} \left\| \tilde{\theta}_{k'}^{\text{OPT}} - \theta_{k'}^{\text{NE}} \right\| + L_c \alpha, \quad (29)$$

where (i) follows from the inequality $\|a - c\| \leq \|a - b\| + \|b - c\|$, (ii) and (iii) follow from Assumption 4, and (iv) follows from the definition of $\tilde{\Theta}_{-k}$. Summing over all k :

$$\sum_k \left\| \tilde{\theta}_k^{\text{OPT}} - \theta_k^{\text{NE}} \right\| \leq L_q (K - 1) \sum_{k'} \left\| \tilde{\theta}_{k'}^{\text{OPT}} - \theta_{k'}^{\text{NE}} \right\| + L_c K \alpha. \quad (30)$$

Rearranging the inequality (assuming $1 - \eta(K - 1) > 0$):

$$\sum_k \left\| \tilde{\theta}_k^{\text{OPT}} - \theta_k^{\text{NE}} \right\| \leq \frac{L_c K \alpha}{1 - L_q (K - 1)}. \quad (31)$$

Consequently,

$$\left\| \tilde{\Theta}^{\text{OPT}} - \Theta^{\text{NE}}(\mathbf{q}) \right\| \leq \sum_k \left\| \tilde{\theta}_k^{\text{OPT}} - \theta_k^{\text{NE}} \right\| \leq \frac{L_c K \alpha}{1 - L_q (K - 1)}. \quad (32)$$

Using Assumption 3:

$$\left\| F(\tilde{\Theta}^{\text{OPT}}) - F(\Theta^{\text{NE}}(\mathbf{q})) \right\| \leq L \cdot \left\| \tilde{\Theta}^{\text{OPT}} - \Theta^{\text{NE}}(\mathbf{q}) \right\| \leq L \cdot \frac{L_c K \alpha}{1 - L_q (K - 1)}. \quad (33)$$

Using Assumption 5:

$$\left\| \Theta^{\text{OPT}} - \tilde{\Theta}^{\text{OPT}} \right\| \leq L_0 \cdot \alpha \Rightarrow \left\| F(\Theta^{\text{OPT}}) - F(\tilde{\Theta}^{\text{OPT}}) \right\| \leq L \cdot L_0 \cdot \alpha. \quad (34)$$

Combining (33) and (34), we obtain the final bound:

$$\left\| F(\Theta^{\text{OPT}}) - F(\Theta^{\text{NE}}(\mathbf{q})) \right\| \leq \alpha L \left(L_0 + \frac{L_c K}{1 - L_q (K - 1)} \right). \quad (35)$$

□

Theory vs. Practice. This bound aligns with the intuition that the NE solution approaches the global optimum as $\alpha \rightarrow 0$ (i.e., when all clients care only about global accuracy and ignore local privacy costs, as in standard VFL), highlighting the practicality of our distributed algorithm in real-world scenarios, where clients not only benefit from global accuracy gains through collaboration but also incur local privacy costs that must be carefully balanced without a global view.

7.4 Sampling-based Approximation of q_k to Address Potential Scalability Issues

Most existing VFL studies [1, 45, 21] suggest that the number of clients (K) is typically small in practice, as the number of splittable features or modalities corresponding to the same batch of samples is limited in the real world. As a result, scalability issues are relatively minor compared to the accuracy–privacy trade-off in VFL. However, we still provide a sampling-based approximation for q_k to address potential scalability issues in extreme cases where K is very large, although such cases are rare and impractical in real-world VFL settings. This sampling-based approximation reduces the computational burden of calculating the exact values of q_k ; however, we do not claim this efficiency improvement as our core contribution, as it is orthogonal to our main objective of balancing the trade-off between global accuracy and local privacy from an equilibrium perspective.

For any coalition $S \subseteq [K] \setminus \{k\}$, the marginal contribution of adding client k to S is defined as $\hat{u}(\Theta[S \cup \{k\}]) - \hat{u}(\Theta[S])$, where $\hat{u}(\Theta[S])$ denotes the normalized accuracy utility of coalition S . The decomposition parameter q_k is defined as: $q_k = \sum_{S \subseteq [K] \setminus \{k\}} \frac{|S|!(K-|S|-1)!}{K!} \cdot (\hat{u}(\Theta[S \cup \{k\}]) - \hat{u}(\Theta[S]))$. Computing the exact value of q_k requires evaluating the marginal contribution for all 2^{K-1} possible subsets $S \subseteq [K] \setminus \{k\}$, resulting in a complexity of $\mathcal{O}(2^K)$. To reduce this, we

introduce a sampling-based approximation. Instead of evaluating all subsets, we randomly sample permutations of clients and estimate the marginal contributions. Specifically, we adopt a Monte Carlo estimation for q_k by sampling from a uniform distribution $\mathcal{M}$ over permutations of the $K - 1$ clients. Let P_k^δ denote the set of clients appearing before k in permutation δ . The approximate value $\hat{q}_k$ is computed as: $\hat{q}_k = \frac{1}{|\mathcal{M}|} \sum_{\delta \in \mathcal{M}} (\hat{u}(\Theta[P_k^\delta \cup \{k\}]) - \hat{u}(\Theta[P_k^\delta]))$, where $\mathcal{M}$ denotes the set of sampled permutations. Empirically, we set $|\mathcal{M}| = K \cdot \log K \cdot \frac{1}{\epsilon}$, where ϵ is a small constant that balances approximation accuracy and computational cost. This approximation reduces the complexity from $\mathcal{O}(2^K)$ to $\mathcal{O}(|\mathcal{M}|)$, offering a significant improvement in computational efficiency.

In sum, we offer an optional solution to further reduce the complexity of computing q_k and improve computational efficiency. However, we do not claim this efficiency improvement as our core contribution, as this paper focuses on the accuracy-privacy trade-off, and the efficiency-related strategy is orthogonal to our main objective. Besides, we highlight that there is no additional computational overhead during the inference phase, which makes it practical for most VFL applications [1, 51]. For example, in collaborative scenarios involving healthcare (e.g., different hospitals) or finance (e.g., banks and FinTech companies), the importance of privacy outweighs that of training efficiency.

7.5 Implementation and Reproducibility

We implement our framework using PyTorch and run all experiments on NVIDIA GeForce RTX 4090 GPUs. We use three publicly available datasets with different application scenarios in our experiments. We adopt the F1 score instead of accuracy as the metric for some imbalanced datasets. In the following, $\uparrow$ and $\downarrow$ indicate that higher or lower values of the respective metrics correspond to better performance.

For the *IEMOCAP* dataset¹, we follow the preprocessing steps in [52], partitioning the data across three modalities for $K = 3$ clients. The client with visual data trains a ResNet18 with a penultimate layer, while the clients with text or audio data train LSTMs with linear layers. The concatenated embeddings are fed into the server head model, which consists of two fully connected layers. The surrogate head model (adversary classifier) used for local adversarial training consists of a fully connected layer, which may differ across clients due to varying feature dimensions. We use $u(x) = \frac{10}{e^x}$ as the accuracy utility function and $c_k(x) = \frac{k}{e^x + 1}$ as the privacy cost function, where k is a constant that can be set by each client. Notably, these functions are predefined by the problem and not designed by our algorithm; their role is simply to convert different losses into meaningful values for joint (global) accuracy and (local) privacy training. We perform 5-fold cross-validation for hyperparameter selection, using a grid search over η in [0.0001, 0.005]. We train for 200 epochs with a batch size of 64. Gender prediction F1 score ($\downarrow$) is used as the privacy task metric, while multi-emotion recognition weighted F1 score ($\uparrow$) is used to evaluate the target task performance.

For the *MIMIC-III* dataset², we follow the preprocessing steps in [21] and align samples for different tasks. We then vertically partition the data along the 76-feature axis for $K = 2$ clients (later increased to 4 or 19 for scalability experiments in Section 5.4). Each client trains a local LSTM with a linear layer. The concatenated embeddings are then fed into the head model on the server, which consists of a fully connected layer. The surrogate head model (adversary classifier) used for local adversarial training also consists of a fully connected layer, which may differ across clients due to varying feature dimensions. We use $u(x) = \frac{10}{e^x}$ as the accuracy utility function and $c_k(x) = \frac{k}{e^x + 1}$ as the privacy cost function, where k is a constant that can be set by each client. Notably, these functions are predefined by the problem and not designed by our algorithm; their role is simply to convert different losses into meaningful values for joint (global) accuracy and (local) privacy training. We perform 5-fold

Algorithm 1: NASHCODER

Initialize: $\theta_0^0, \{\theta_k^0\}_{k=1}^K, \{g_k^0\}_{k=1}^K$
for $t = 0, 1, \dots, T - 1$ **do**
 if $t \bmod Q = 0$ **then**
 for $k = 1, \dots, K$ **in parallel do**
 Send $h_k(\theta_k^t; \mathbf{x}_k^B)$ to server;
 $\tilde{\Phi}^t \leftarrow \{h_1(\theta_1^t), \dots, h_K(\theta_K^t)\}$;
 Update $\{q_k\}_{k=1}^K$;
 Server sends $\tilde{\Phi}^t$ and $\tilde{\theta}_0^t$ to clients;
 else
 for *server* ($k = 0$) **do**
 Update θ_0^t using $\{\tilde{\Phi}^t; \mathbf{y}^B\}$;
 for $k = 1, \dots, K$ **in parallel do**
 Update (see Eq. in main text):
 $g_k^t \leftarrow \arg \min \text{Eq. (4)}$;
 $\theta_k^t \leftarrow \arg \min \text{Eq. (5)}$;

¹<https://sail.usc.edu/iemocap/>

²<https://mimic.mit.edu/>

cross-validation for hyperparameter selection, using a grid search over η in [0.001, 0.02]. We train for 400 epochs with a batch size of 256. The F1 score ($\downarrow$) of LOS>2 prediction serves as the privacy task metric, while the F1 score ($\uparrow$) evaluates the target task performance, given that only 16% of samples are positive (i.e., most patients did not experience in-hospital mortality).

For the *CelebA* dataset³, we follow the preprocessing steps in [53]. Each image is vertically split into two parts along the central line for $K = 2$ clients. Each client trains a ResNet18 model [54] with a penultimate layer, and the concatenated embeddings are fed into the server head model, which consists of three fully connected layers. The surrogate head model (adversary classifier) used for local adversarial training consists of two fully connected layers, which may vary across clients. We use $u(x) = \frac{10}{e^x}$ as the accuracy utility function and $c_k(x) = \frac{k}{e^x + 1}$ as the privacy cost function, where k is a constant configurable by each client. Notably, these functions are predefined by the problem and not designed by our algorithm; their purpose is to convert different loss values into meaningful quantities for joint (global) accuracy and (local) privacy training. We perform 5-fold cross-validation for hyperparameter selection, using grid search over η in the range [0.0001, 0.002]. We train for 150 epochs with a batch size of 64. Due to the wide availability of attributes in CelebA (40 binary attributes per facial image), we use the prediction accuracy ($\downarrow$) for “male” and “young” as metrics to evaluate heterogeneous privacy task performance. Additionally, we use accuracy ($\uparrow$) for predicting “smiling” and “gray hair” as metrics for two different target tasks, providing a broader evaluation of generalizability and applicability.

Commonly Used Privacy Metrics. Our privacy performance metrics (accuracy or F1 score), from the perspective of a potential attribute inference attacker (surrogate model), align with those used in several adversarial learning-based privacy-enhancing methods in non-FL contexts [8, 9, 19, 55], and are also commonly adopted in studies on attribute inference attacks [6–9]. Notably, our goal is to address new challenges in the VFL setting, such as heterogeneous privacy, the trade-off between global accuracy and local privacy, and distributed optimization without a global view, as discussed in Section 1 of the main paper. We fairly evaluate our method’s performance against five baselines in terms of the accuracy-privacy trade-off, using these standard metrics for attribute inference attacks, which are the primary focus of this paper.

Implementation of Baselines: We implement five baselines with the following configurations:

(1) *Standard VFL*⁴ transmits embeddings instead of feature-partitioned raw data, without additional protection for the embeddings, making them vulnerable to attribute inference attacks [6–9].

(2) *DP-Gaussian* applies the differential privacy (DP) strategy in VFL by adding Gaussian noise to protect privacy [49, 11, 12, 48]. Specifically, in DP-Gaussian, noise is sampled from a Gaussian distribution $\mathcal{N}(0, \sigma^2)$, where σ is the standard deviation. The parameter σ determines the magnitude of the noise and directly affects the privacy budget ϵ in (ϵ, δ) -DP. According to the Gaussian mechanism in DP, the added noise satisfies: $\sigma \geq \frac{\Delta_f \sqrt{2 \ln(1.25/\delta)}}{\epsilon}$, where Δ_f is the sensitivity of the function f , ϵ is the privacy budget, and δ is the failure probability. Sensitivity Δ_f measures the maximum change in the function output caused by altering a single input. For embedding-based models, Δ_f can be derived from the norm of the embeddings. The privacy budget ϵ controls the strength of privacy guarantees, with smaller ϵ providing stronger privacy. The failure probability δ represents the probability that the privacy guarantee may be violated, typically set to a very small value. While the theoretical formula provides a lower bound for σ , the actual value is often adjusted experimentally. The range [0.01, 0.5] is explored to balance privacy and utility. Increasing σ improves privacy by making the embeddings less distinguishable but may lead to reduced target task performance due to excessive noise. Conversely, reducing σ enhances accuracy but increases the risk of privacy leakage, particularly under sensitive attribute inference attacks.

(3) *DP-Laplacian* employs a different DP strategy by injecting Laplacian noise to safeguard information [49, 10, 50]. For DP-Laplacian, noise is sampled from a Laplacian distribution with zero mean and scale parameter b , where b determines the magnitude of the added noise. The privacy guarantees in $(\epsilon, 0)$ -DP depend on b . The scale parameter b is calculated as: $b = \frac{\Delta_f}{\epsilon}$, where Δ_f is the sensitivity of the function, and ϵ is the privacy budget. Similar to the Gaussian case, Δ_f represents

³<https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html>

⁴There are two lines of VFL: one assigns an active party (label holder) and passive parties, transmitting partial gradients; the other, which we adopt, transmits embeddings and is widely adopted in recent VFL [21, 22, 5].

the maximum change in the function output when a single input changes. In VFL scenarios, this could relate to the maximum difference between two embedding vectors. A smaller privacy budget ϵ provides stronger privacy but requires a larger b , resulting in more noise. The parameter b is tuned experimentally within the range $[0.007, 0.35]$. Increasing b enhances privacy by making the embeddings more obfuscated but can negatively affect model performance due to excessive noise. Conversely, decreasing b retains better utility but weakens the privacy guarantees, particularly when dealing with sensitive information.

(4) *C-VFL* adopts a compression-based privacy-enhancing strategy by pruning the embeddings [21]. Similar to the original implementation, we primarily use top- k sparsification, the sparsification level is determined by the sparsity ratio k , which specifies the proportion of retained values relative to the total number of embedding elements. This approach reduces the volume of information transmitted, thereby mitigating potential privacy risks during communication. By discarding less significant embedding components, it becomes harder for adversaries to reconstruct sensitive information from pruned embeddings. To evaluate the trade-off between privacy and utility, we experiment with k values in the range $[0.1, 0.5]$. A lower sparsity ratio k enhances privacy by reducing the information shared, but it may degrade model performance due to the loss of essential features for downstream tasks. Conversely, a higher sparsity ratio retains more useful information, improving downstream accuracy at the cost of increased privacy risks. Careful tuning of k is essential to achieve an optimal balance between privacy and utility. Moreover, embedding pruning in C-VFL inherently supports scalability by reducing the size of the transmitted embeddings, making it well-suited for large-scale VFL scenarios with limited communication bandwidth.

However, as discussed in our main paper, such efficiency-related strategies are orthogonal to our work, since our main objective is to balance the trade-off between global accuracy and local privacy, and to design efficient algorithms for distributed optimization without a global view.

(5) *Ours with uniform decomposition*, i.e., $q_k = \frac{1}{K}, \forall k$, serves as an important baseline to highlight the advantages of our adaptive decomposition strategy, which builds on the same idea of selectively protecting client-specific sensitive attributes. Our goal is to construct a better equilibrium model by reshaping the value functions in Eq. (2) (see Eq. in main paper), compared to uniform decomposition, thereby achieving improved overall outcomes in terms of the accuracy-privacy trade-off.

7.6 Preliminary Experimental Results on Baselines

Since most privacy-enhanced baselines focus on adjusting the privacy level (privacy budget) rather than maximizing the value function (i.e., the difference between accuracy utility and privacy cost) as in our formulation in Section 3, we conduct extensive preliminary experiments to tune their privacy settings in order to ensure a fair evaluation of the baselines. The preliminary results on the IEMOCAP dataset are shown in Figs. 1 to 3, the results on the MIMIC-III dataset are shown in Figs. 4 to 6, and the results on the CelebA dataset are shown in Figs. 7 to 9. For each dataset and each α , we select the privacy setting (indicated by the red point) that yields the highest normalized value to ensure a fair comparison with our method, as it aligns with the setting described in Section 3.

Notably, we perform the following preliminary experiments with different trade-off parameters α to observe the overall accuracy-privacy trade-off trend, as shown in Figs. 5 to 7, since evaluating improvement based on a single trade-off point would be unclear given the two-dimensional metrics.

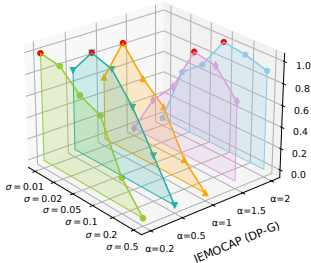


Figure 1: Normalized values under varying σ (IEMOCAP).

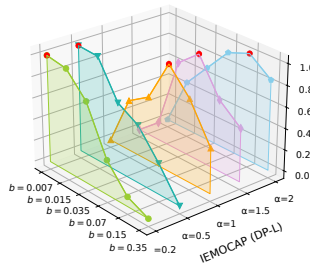


Figure 2: Normalized values under varying b (IEMOCAP).

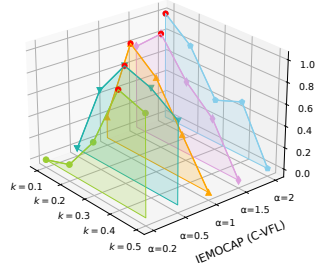


Figure 3: Normalized values under varying k (IEMOCAP).

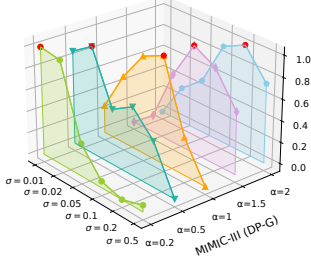


Figure 4: Normalized values under varying σ (MIMIC3).

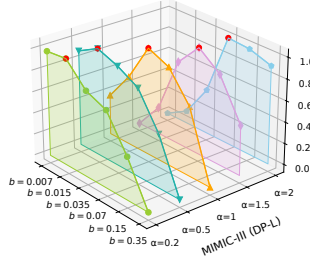


Figure 5: Normalized values under varying b (MIMIC3).

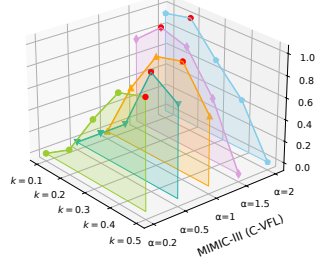


Figure 6: Normalized values under varying k (MIMIC3).

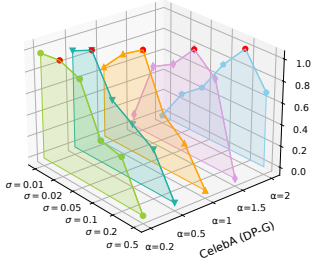


Figure 7: Normalized values under varying σ (CelebA).

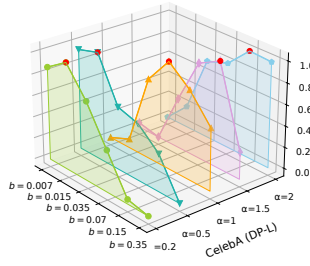


Figure 8: Normalized values under varying b (CelebA).

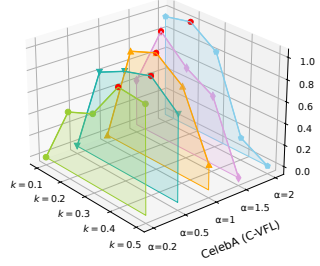


Figure 9: Normalized values under varying k (CelebA).

7.7 Additional Experimental Results

In this section, we present additional experimental results to further supplement the experiments and ablation studies reported in our main paper.

Impact of surrogate head architecture. As discussed in the threat model in Section 3, in practice, clients lack knowledge of the attacker’s model and only know which attributes (e.g., gender or race) are sensitive to them. As part of our approach, each client incorporates a surrogate head g_k to approximate the behavior of a potential attribute inference attacker. We therefore evaluate the impact of different architectures for g_k , although they are usually implemented as shallow adversarial classifiers. This is because a more complex surrogate model does not necessarily yield better performance in practice, as its effectiveness also depends on factors such as feature dimensionality and sample size. We follow the architectural design of surrogate models used in several adversarial learning-based privacy studies in non-FL contexts [8, 9, 19, 20]. Specifically, we consider surrogate heads consisting of one or two fully connected layers (denoted as S1 and S2) and assign them to two clients to simulate three different combinations. We present results on the MIMIC-III dataset, with similar observations on the other two datasets. As illustrated in Fig. 10, *NashCoder* defends against attribute inference attacks regardless of the surrogate head architecture, which aligns with our motivation to locally obfuscate (client-specific) sensitive attributes via adversarial training for robust privacy protection.

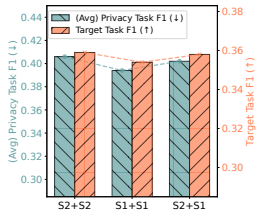


Figure 10: Impact of g_k .

Imbalanced feature partitioning. Since VFL involves partitioning the feature space, it may be difficult to ensure a balanced distribution, leading to imbalanced feature partitioning in real-world scenarios [56–58]. Although this issue is not highlighted in our main paper, our framework effectively addresses it because the global objective is adaptively decomposed based on each client’s contribution, as described in Section 4.3.



Figure 11: Vertical split (symmetrical case) mimics relatively balanced feature partitioning.

We conduct additional experiments on the CelebA dataset and follow [56] to introduce feature imbalance, which refers to the unequal importance of features across different clients. Specifically, we split the facial images using two partitioning schemes: a symmetrical vertical split, where each face image is divided vertically into left and right halves, resulting in a relatively balanced distribution of facial features, as shown in Fig. 11; and an asymmetrical horizontal split, where each face image is divided horizontally into upper and lower halves, leading to a more asymmetric distribution of informative features, as shown in Fig. 12. Note that although this is a horizontal split, it still follows the VFL principle because it involves feature partitioning.



Figure 12: An asymmetrical horizontal split mimics relatively imbalanced feature partitioning.

As shown in Fig. 13, the asymmetrical split results in poorer performance compared to the symmetrical split, which is intuitive since imbalanced feature importance complicates joint training when all clients are assigned equal weights, potentially leading to degraded learning performance. However, our method still outperforms other baselines in such scenarios due to our adaptive decomposition strategy, which dynamically adjusts the decomposition weights $\{q_k\}$ for each client k based on their past contributions from an equilibrium perspective.

Importantly, we use the Shapley value to quantify each client’s marginal contribution, making the decomposition process privacy-cost-agnostic and based solely on each client’s feature importance (marginal contribution) for the global target task. This enables our equilibrium-driven VFL framework to handle imbalanced feature partitioning effectively by performing distributed optimization in a non-uniform yet fair manner.

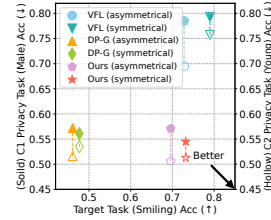


Figure 13: Results under imbalanced partitioning.

7.8 Additional Discussion

Knowledge of sensitive attributes. Each client can flexibly define its own sensitive attributes (e.g., gender or race) locally, and this definition remains confidential to other parties, which aligns with real-world scenarios where improving local privacy requires at least knowing the privacy target. In our experiments, even when sensitive attributes are homogeneous across clients, each client is unaware of this homogeneity. Most importantly, c_k may still differ across clients.

In other words, privacy is controlled locally while accuracy is achieved globally, which is consistent with the VFL setting in practical applications, where conflicting objectives arise not only within each client but also across different clients, thereby posing challenges in addressing heterogeneous privacy and balancing the trade-off during distributed optimization without a global view.

Label access. There are two lines of VFL: one assigns an active party (label holder) and passive parties that transmit partial gradients, while the other, adopted in this work and widely used in recent VFL studies, transmits embeddings to enable flexible feature space partitioning and support arbitrary head models for feature fusion. In this setting, target task labels are assumed to be accessible to all parties, following a commonly used assumption in many recent VFL studies [21, 2, 22, 5]. In future work, we will investigate the applicability of our framework in semi-supervised and unsupervised learning scenarios, which are orthogonal to this paper’s focus on the accuracy-privacy trade-off.

Generalizability to other attacks. Although our work focuses on attribute inference attacks [6–9], which constitute a primary form of privacy leakage on valuable embeddings, the threat model described in Section 3 can be extended to other types of attacks, such as feature reconstruction attacks [59–62]. Specifically, the surrogate model can be adapted from an adversarial classifier to an adversarial reconstructor. In this case, adversarial training can be performed using a reconstruction loss instead of cross-entropy loss, and the evaluation metrics can be modified accordingly to incorporate similarity-based measures such as the Structural Similarity Index Measure (SSIM) or Peak Signal-to-Noise Ratio (PSNR). Nevertheless, a fundamental challenge remains in balancing the trade-off between global accuracy

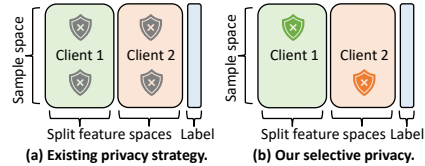


Figure 14: Unlike one-size-fits-all privacy, we selectively obfuscate sensitive attributes without affecting useful information, thus mitigating accuracy loss.

and local privacy in a distributed setting. Our equilibrium-driven VFL framework with selective privacy protection can still be applied in this context.

Applications. Our work is directly applicable to various VFL scenarios [1] involving heterogeneous privacy and accuracy-privacy trade-offs, which are common and critical challenges in real-world applications. These include: (1) healthcare applications for collaborative diagnostics, where multiple hospitals possess different types of medical records with distinct privacy objectives corresponding to the same patients; (2) financial applications, where banks and FinTech companies maintain separate financial records with different sensitive attributes for the same customers and collaborate on loan approvals; and (3) other multimodal scenarios, such as emotion recognition and human action recognition, where certain attributes (e.g., gender or race) are considered sensitive for local clients and are unrelated to the target task, and thus should not be accurately inferred by potential attackers.

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